

Audit Report

(Join Cohort)

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1. Executive Summary

This report presents a security assessment of Join-cohort cloud-hosted static website deployed using Amazon Web Services (AWS). The environment was designed using Amazon S3 for static website hosting, Amazon CloudFront for secure content delivery and HTTPS support, and Amazon Route 53 for DNS management and traffic routing.

The purpose of this assessment was to evaluate the security posture of the deployed environment, identify potential risks, and recommend mitigation strategies aligned with cloud security best practices and governance principles following NIST SP 800-53 security control concepts.

The assessment identified several cloud security considerations related to public access configuration, access control management, logging visibility, and web application protection. Recommendations were provided to improve monitoring, strengthen access management, and reduce the overall attack surface of the environment.

This project provided practical exposure to cloud security analysis, risk assessment methodologies, governance-focused documentation, and mitigation planning within AWS environments.

2. Scope of Assessment

The following AWS services and configurations were reviewed during this assessment:

Amazon S3

Amazon CloudFront

Amazon Route 53

The assessment focused on:

Public exposure risks

Security best practices

Monitoring and logging considerations

Governance and risk mitigation recommendations

2.1 Assessment Methodology

The assessment was conducted using a qualitative security review approach informed by cloud security best practices and security control concepts inspired by NIST SP 800-53.

The review considered:

- Audit logging and monitoring visibility
- System and communications protection
- Secure cloud configuration
- Risk identification and mitigation planning

3. Architecture Overview

The static website architecture follows the structure below:

User → Route 53 → CloudFront → Amazon S3

Amazon S3 used to host the static website files including HTML, CSS, and image assets.

Amazon Cloudfront used as a Content Delivery Network (CDN) to provide HTTPS delivery, caching, and improved performance.

Amazon Route 53 used for DNS management and routing the custom domain to the CloudFront distribution.

4. Security Risk Assessment

4.1 Amazon S3 Security Review

NIST SP 800-53 Access Control (AC)

NIST SP 800-53 Configuration Management (CM) and Risk Assessment (RA)

Observation

The S3 bucket hosting the website required public access configuration to allow website content delivery.

Security Considerations

- Public bucket access can increase exposure to unauthorized access.
- Misconfigured bucket policies may expose sensitive files if not properly controlled.
- Static website hosting environments should avoid storing confidential information within publicly accessible buckets.

Best Practices Applied

- Bucket policies were configured specifically for website hosting requirements.
- Sensitive information was not stored within the bucket.
- CloudFront was used to improve secure content delivery.

4.2 CloudFront Security Review

NIST SP 800-53 System and Communications Protection (SC)

Observation

CloudFront was configured to distribute website content securely through HTTPS.

Security Benefits

- HTTPS encryption improves secure communication between users and the website.
- CloudFront helps reduce direct exposure of the S3 bucket.
- CDN caching improves availability and performance.

Additional Recommendations

- Implement AWS WAF for additional web application protection.
- Enable detailed access logging for improved monitoring visibility.

4.3 Route 53 Security Review

Observation

Route 53 was configured for DNS management and routing traffic to CloudFront.

Security Considerations

- Proper DNS configuration helps ensure secure and reliable routing.
- DNS misconfiguration may impact availability or redirect traffic incorrectly.

Best Practices Applied

- CloudFront alias records were used for domain routing.
- Secure HTTPS delivery was configured through CloudFront.

5. Identified Risks and Mitigation Strategies

Risk Identified	Potential Impact	Likelihood	Mitigation Strategy
Public S3 Bucket Exposure	Unauthorized public access to files	Medium	Restrict access through CloudFront and apply least privilege policies
Missing AWS WAF Protection	Increased exposure to web-based attacks	Medium	Implement AWS WAF with managed security rules
Lack of CloudTrail Logging	Reduced visibility into account activity	High	Enable AWS CloudTrail monitoring

6. Recommendations

The following recommendations are suggested to improve the overall security posture of Join Cohort website environment:

- Enable AWS CloudTrail logging for account activity monitoring.
- Implement AWS WAF to improve web application protection.
- Enable S3 versioning for improved recovery capabilities.
- Regularly review bucket policies and public access configurations.
- Monitor CloudFront and AWS service logs for unusual activity.
- Continue documenting security decisions and mitigation strategies.

7. Lessons Learned

- AWS cloud architecture design
- Cloud security considerations
- Risk identification and mitigation planning
- Governance-focused security documentation
- DNS and HTTPS configuration
- Security best practices for static website hosting

8. Conclusion

This project demonstrated the deployment and security assessment of a cloud-hosted static website architecture using AWS services including Amazon S3, CloudFront, and Route 53.

The assessment identified key cloud security considerations and provided mitigation recommendations aligned with security best practices and governance security principles following NIST SP 800-53.

Overall, the project provided valuable hands-on experience in cloud security analysis, risk management, and security-focused documentation within AWS environments.